

When Verification Axes Can Support Scientific Claims: Identifiability, Attainability, and Non-Compensatory Promotion

Anonymous authors

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Abstract

A benchmark score can fail to support a scientific-competence claim because the target is not identifiable from the intended evidence, because a compared system cannot emit the target terminal, or because the panel is saturated. We give a verification-axis theory that separates these failures. For finite benchmark distributions, the optimal exact-terminal risk is the fibrewise Bayes error of the representation available to the decision rule. Zero risk is attainable if and only if the gold terminal is constant on every positive-probability representation fibre and belongs to the rule’s output alphabet. The same result yields a donor-product factorization theorem: a composition of provenance, verification, custody and authorization mechanisms can reproduce a scientific-promotion relation exactly if and only if its exposed state is target-sufficient; otherwise its mixed fibres impose an irreducible error bound. A complementary total-variation bound shows when the observable record cannot distinguish two opposite interpretations of a competence result.

We apply this theory to three prospectively separated evidence layers. In protected V2, a non-compensatory authority pipeline produced 0/360 false promotions versus 180/360 for the strongest frozen mechanism proxy while both promoted 60/60 clean positives. The registered H3 abstention contrast remains NOTSUPPORTED: every system scored 30/30 because an empty evidence list made the family trivial. A distinct V3 repair cleared the frozen fourteen-probe register on the exact CANNOTCHECK axis and produced 30/30 versus 0/30 for the V2-selected comparator and 15/30 for the one comparator able to escalate. Because nine of ten comparators cannot emit CANNOTCHECK, that result establishes terminal expressiveness, not general epistemic judgement. Finally, the separate P4-X successor grants a strong product donor-complete provenance, verification, artifact/version, custody, epoch and generic-authorization mechanisms. Across 400 frozen exact contracts, the non-compensatory scientific-promotion relation was correct on 400/400 cases versus 250/400 for that product and 50/400 for a compensatory product; an information-equivalent typed product tied at 400/400. The paired improvement over the strong product was 0.375 (domain-stratified bootstrap 95% CI [0.3275, 0.4225]), with zero false promotions and complete clean promotion. These results support the factorization boundary and a scientific-promotion interface on the exact contracts, not naturalistic, provider-level, or universal superiority. A frozen paired 768-cluster naturalistic successor is reported as a prospective design only; no external-panel outcome is claimed.

Keywords: scientific verification; benchmark identifiability; attainable risk; provenance; authorization; protected evaluation; abstention.

1 Introduction

Language-model research systems increasingly retrieve sources, synthesize claims, attach citations, run automated verification, and decide whether outputs may be acted upon. This creates two coupled scientific

problems. First, a claim may be correct but attributed to the wrong source; a source may support a nearby claim but not the assigned one; a checker may be non-empty yet non-discriminating or share lineage with the answer; and an evaluator may be contaminated or tampered with after the fact. In each case, a scalar confidence score can remain high even though a prerequisite for scientific authority is absent. Second, a benchmark may report a clean number without measuring the intended competence: its gold terminal may be exposed by construction artifacts, unavailable in a comparator’s interface, or constant over the entire panel.

We study a stricter object: a *scientific-authority transition*. A proposal can be plausible and well supported but is promoted only when a registered set of content, provenance, checker, evaluator, scope, dependence, and freshness obligations all pass. Unresolved prerequisites do not contribute a low score; they block escalation. This design is motivated by source-aware verification, multi-source attribution, claim-level auditability, evidence influence, iterative fact checking, evaluator-tampering benchmarks, search-time contamination, partial-evidence awareness, and evidence-backed authorization (Alvarez et al., 2026; Li et al., 2024; Rasheed et al., 2026; Faizan & Alharthi, 2026; Xie et al., 2024; Atinafu & Cohen, 2026; Wang et al., 2026c; Sadeghi et al., 2026; Tallam, 2026; Wang et al., 2026b; Zhang et al., 2026; Llambí-Morillas & Fernández-Fernández, 2026). Recent academic-integrity and agentic-abstention benchmarks also make explicit that declining to complete an unsupported action can itself be the correct behavior rather than a failure to optimize task success (Yang et al., 2026; Liu et al., 2026).

The contribution moves above those component techniques rather than relabeling them. We ask when a verification axis can support a scientific claim at all, and when donor-complete local mechanisms jointly suffice for target scientific promotion. Section 3 gives an exact attainability criterion, a mixed-fibre lower bound, a donor-product factorization theorem, and a diagnostic separation among construction saturation, nuisance leakage, interface limitation, and decision failure. The empirical studies then ask whether composing the donor mechanisms as a *non-compensatory, protected authority transition* reduces false scientific promotion while preserving clean promotion. Throughout, the pipeline built to that rule is called the *governed pipeline*, so that each claim reads as a claim about the rule rather than about a product; Section 5 names the implementation that instantiates it.

The V2 study was executed after the subject, comparator/ablation mechanisms, evaluator, statistical rules, and split-generation code were frozen. A first 420-case protected campaign on an older subject remains valid evidence for that older revision but does not authorize the repaired one. A separate 39-case live-model arm remains exploratory because post-run audit found label/denominator inconsistencies and a hidden-family execution leak. We therefore generated a fresh post-repair hidden split and executed the publication-authorizing V2 campaign without reusing either prior outcome for tuning. V3 and P4-X are separately identified successors; neither is pooled with V2 or used to rewrite its terminals.

Our contributions are:

1. a general finite-space theory of verification-axis attainability, including the exact Bayes-risk expression, a necessary-and-sufficient zero-risk condition, a total-variation bound for wider claim identifiability, a nuisance-identifiability criterion, and a donor-product factorization boundary;
2. a protected V2 result of 0/360 versus 180/360 false promotions with 60/60 clean promotion in both arms, together with the retained negative finding that H3 was non-discriminating because its construction was saturated;
3. an exact-axis V3 repair that clears the frozen shortcut probes and obtains 30/30 correct CANNOTCHECK terminals versus 0/30 and 15/30 for the relevant comparators, while explicitly identifying that result as interface expressiveness rather than general epistemic judgement; and
4. P4-X, a donor-engulfing successor in which the non-compensatory scientific-promotion relation scores 400/400 versus 250/400 for a strong generic-authorization product and 50/400 for a compensatory product, while an information-equivalent typed product ties at 400/400 as the factorization theorem requires.

The widest empirical target—naturalistic, cross-domain scientific-promotion judgement against CANNOTCHECK-capable external comparators—remains prospective. Its frozen design contains 768 paired source-family clusters across four domains and eight unidentifiability mechanisms, with an independently held external panel and source-disjoint replication required before any such claim can be promoted.

2 Related work

Source ownership and attribution. Attribution evaluation across multiple candidate sources is an established measurement problem (Li et al., 2024); source-aware factuality work targets cross-source conflation directly (Alvarez et al., 2026); and scientific claim–evidence benchmarks test whether retrieved evidence supports or contradicts a claim (Javaji et al., 2025). Claim-source retrieval sharpens the same point from the retrieval side, since recovering the source a scientific statement actually came from is a different task from finding a source that would support it (Schreieder et al., 2026). Together these establish claim correctness, source ownership, and semantic support as separate coordinates, which is the premise this paper builds on.

Provenance-sensitive authorization. A current controlled audit makes the authorization distinction especially direct: evidence can be relevant to an agent decision without being authorized to determine that decision, and target-specific tests can hold the task and proposition fixed while varying only source authority (Liao, 2026). The relevance-versus-authorized-evidence distinction is therefore already established. What remains open is the empirical question studied here: whether a protected conjunction of content, ownership, support, checker, influence, evaluator, access, and contamination prerequisites behaves differently from its strongest single component when both meet the same hostile battery.

Partial evidence and formal authorization. The donor frontier is stronger than provenance-sensitive factuality alone. Partial Evidence Bench studies unsafe completeness and structured gap reporting when relevant evidence lies outside an authorized view (Tallam, 2026). AuthGraph aligns execution provenance with an independently maintained authorization graph (Wang et al., 2026b); FAVA lowers natural-language constraints into evidence-backed permission graphs with deterministic authorization (Zhang et al., 2026); and CVA binds principal, request, context, and policy satisfaction in a cryptographically verifiable authorization object (Llambí-Morillas & Fernández-Fernández, 2026). P4 treats all four as donor-owned advances. Its residual question is not whether evidence can be partial, provenance can be aligned with permission, or authorization can be formal. It is whether the joint donor state is sufficient for the target *scientific-promotion* terminal. Section 3 states the exact factorization condition: once a donor product exposes a target-sufficient state and the same terminal relation, it is extensionally equivalent.

Auditability, citation fidelity, and influence. Claim-level auditability work argues that research-agent outputs require persistent semantic provenance rather than only fluent text (Rasheed et al., 2026). ProvenAI separates citation fidelity from whether cited evidence influenced the answer path (Faizan & Alharthi, 2026). We adopt that separation: citation presence cannot substitute for evidence-use provenance.

Research-integrity provenance and declared behavior. INSPECT-AI and RIPE-KG represent the provenance of research-integrity assessments over clinical-trial publications, showing a current scientific setting in which the assessment process itself is an auditable object (Markovic et al., 2026). Behavioral Integrity Verification formalizes a complementary declared-versus-actual integrity problem for agent skills (Wu et al., 2026), and certified speculative execution shows the same instinct applied to untrusted agent actions: let the work proceed, but hold its effects behind a check that must clear before they become authoritative (Zhou et al., 2026). These are parent-domain precedents for explicit provenance and contract checking.

Iterative verification and evidence escalation. FIRE iterates retrieval and verification rather than treating a single retrieval pass as final (Xie et al., 2024). DeepSciVerify motivates escalation from an abstract-level judgment to fuller evidence when initial evidence is insufficient (Sadeghi et al., 2026). Stage-wise fact-checking benchmarks make the complementary measurement argument, that an end-to-end verdict hides which stage actually failed (Lin et al., 2026). The first two mechanisms inform frozen comparator proxies

here, and the third informs the decision to report per-family and per-coordinate outcomes rather than one aggregate.

Evaluator integrity, benchmark auditing, and contamination. RewardHackingAgents treats evaluator manipulation and held-out leakage as an explicit benchmark dimension (Atinafu & Cohen, 2026). Search-time contamination work shows that browsing systems can retrieve public benchmark answers during inference (Wang et al., 2026c). BenchGuard and Automated Benchmark Auditing show that benchmark specifications, environments, ground truths, and grading logic can themselves be defective (Tu et al., 2026; Wang et al., 2026a). The Holistic Agent Leaderboard further demonstrates the value of standardized harnessing and trace inspection, including direct observation of agents searching for benchmark material rather than solving the intended task (Kapoor et al., 2025). These results motivate the protected evaluator identity, holdout custody, access telemetry, and independent audit used here.

The verification-axis analysis adds a distinction that benchmark auditing alone does not provide. A task can be free of the registered nuisance shortcuts yet still fail to compare scientific judgement because the target terminal is absent from a system’s interface; conversely, a perfectly expressive interface does not make a mixed information fibre identifiable. Construction audit, information sufficiency, terminal attainability, and observed panel resolution must therefore be reported separately.

Abstention and scientific integrity. SciIntegrity-Bench constructs academic-integrity dilemmas in which honest acknowledgement of inability is the only correct outcome, while AgentAbstain evaluates paired should-act/should-abstain tasks in executable environments (Yang et al., 2026; Liu et al., 2026). They establish abstention as an evaluation target distinct from generic task success. The object measured here is narrower: an undetermined or blocked terminal is authority-specific, produced by an unresolved or failed evidence, checker, or evaluator prerequisite rather than by a general judgement that the task should not be attempted.

3 Verification-axis identifiability and attainability

A benchmark does not measure a scientific competence merely because it assigns a label and reports an accuracy. The label must be recoverable from the intended scientific information, unavailable from construction nuisances, expressible by the systems being compared, and capable of varying over the comparison panel. These are different requirements. We formalize them before interpreting the protected results.

Let $Y_a \in \mathcal{T}$ be the correct terminal on verification axis a for a random benchmark unit, let D be the representation available to a decision rule, and let $\mathcal{A} \subseteq \mathcal{T}$ be the terminals that the rule is allowed to emit. For a finite benchmark distribution, the smallest attainable exact-terminal error is

$$R_a^*(D, \mathcal{A}) = 1 - \mathbb{E}_D \left[\max_{t \in \mathcal{A}} \Pr(Y_a = t \mid D) \right]. \quad (1)$$

This quantity separates a poor implementation from a task that its information or interface makes impossible.

Proposition 1 (exact axis attainability). There exists a decision rule $g(D) \in \mathcal{A}$ with zero exact-terminal error if and only if every positive-probability fibre of D contains one target terminal and that terminal belongs to $\mathcal{A}$. Equivalently, $Y_a = g(D)$ almost surely for some g with codomain $\mathcal{A}$.

Proof. On a fibre $D = d$, any rule must emit one terminal. It is correct on the whole fibre exactly when the conditional support of Y_a is the singleton containing that terminal. Summing the best fibrewise choices gives Equation 1, and zero risk gives the stated condition. $\square$

Corollary 1 (donor-product factorization). Let D collect the states exposed by a product of provenance, verification, custody, freshness, and generic-authorization donors, and let Y_{promote} be the target scientific-promotion terminal. An implementation over those donors is extensionally equivalent to the target relation if and only if the target terminal is constant on every donor-state fibre (and belongs to its output

alphabet). If a fibre mixes target terminals, every donor-only rule has error at least

$$\sum_d \Pr(D = d) \left[1 - \max_{t \in \mathcal{A}} \Pr(Y_{\text{promote}} = t \mid D = d) \right]. \quad (2)$$

Thus a separate module has no inherent advantage once a donor product receives a target-sufficient representation and the same decision relation; conversely, no optimization of a representation that merges differently licensed states can recover the missing distinction. This is a factorization boundary, not a claim that scientific authority must be centralized.

Proposition 2 (claim-level indistinguishability). Let W_0 and W_1 be two scientific worlds in which a proposed competence claim has opposite truth values, and let O be the complete observable record available to an adjudicator. Under equal prior weight, every adjudicator has error at least

$$\frac{1 - \text{TV}(\mathcal{L}(O \mid W_0), \mathcal{L}(O \mid W_1))}{2}. \quad (3)$$

In particular, if the observable records have the same distribution, the competence claim is not identifiable and the minimum error is $1/2$.

Proof. For any acceptance region B , the equal-prior error is $\{\Pr_{W_0}(O \in B) + \Pr_{W_1}(O \notin B)\}/2$. Minimizing over B gives one half of one minus the total-variation distance. $\square$

This second bound concerns interpretation rather than terminal prediction. A score can separate output interfaces while leaving two explanations of that score observationally equivalent—for example, a rule that recognizes an epistemic insufficiency and a rule that is merely forced into the only available non-promoting terminal. A broader competence headline remains unidentified until a discriminator changes the observable law of those worlds.

Nuisance identifiability. Let N contain construction-only quantities such as list length, missingness, template, key order, path name, or label-bearing metadata. The nuisance-only Bayes advantage is

$$I_a(N) = \mathbb{E}_N \left[\max_t \Pr(Y_a = t \mid N) \right] - \max_t \Pr(Y_a = t). \quad (4)$$

A scientific interpretation requires the intended semantic representation to make the terminal attainable while nuisance information contributes no material advantage. A finite shortcut register cannot certify $I_a(N) = 0$ for every possible decoder. It certifies only that the frozen probe class found no such advantage, which is why claim-axis authorization must name the probe class and cannot be promoted to whole-register or naturalistic clearance.

Panel resolution and interface attainability. For a frozen panel $\mathcal{H}$ and score M_a , define its observed resolution as

$$\rho_a(\mathcal{H}) = \max_{h \in \mathcal{H}} M_a(h) - \min_{h \in \mathcal{H}} M_a(h).$$

When $\rho_a = 0$, the experiment supplies no comparative ordering on that axis; it does not establish equality of the underlying competences. When $\rho_a > 0$ but some comparators lack a target terminal in their output alphabet, the contrast establishes interface expressiveness before it establishes the finer judgement of when that terminal is scientifically appropriate. A broad verification-axis claim is therefore licensed only inside the intersection of four conditions: semantic attainability, nuisance resistance, matched terminal attainability, and nonzero panel resolution. Failures fall into distinct research programmes rather than one generic null: repair the construction, enlarge the interface, expose a target-sufficient representation, or improve the decision rule.

Application to the three P4 evidence layers. The historical V2 H3 axis had $\rho_a = 0$: an empty evidence list exposed the gold family and every panel member scored 30/30. That result remains NOTSUPPORTED. The later V3 construction removes the registered shortcuts and obtains nonzero resolution, but nine of ten

comparator mechanisms cannot emit CANNOTCHECK; its exact claim is therefore terminal expressiveness under the frozen gate lattice, not general epistemic judgement. The P4-X exact-contract successor tests the donor-factorization boundary more directly. Its ideal typed donor product receives the target-sufficient coordinates and predicate and ties exactly at 400/400, while the registered generic-authorization and compensatory products do not. These layers diagnose different limits and are not pooled.

4 Non-escalating authority transition

Let a proposal contain claim c and evidence references $E = \{e_1, \dots, e_n\}$. Promotion is a conjunction of hard prerequisites rather than a weighted score:

$$G_{\text{content}} : \text{resolved bytes equal the registered evidence digest,} \quad (5)$$

$$G_{\text{owner}} : \text{the assigned source owns the cited content,} \quad (6)$$

$$G_{\text{support}} : \text{the assigned evidence supports } c, \quad (7)$$

$$G_{\text{indep}} : \text{checker lineage is admissibly independent,} \quad (8)$$

$$G_{\text{disc}} : \text{checker rejects the frozen hostile battery,} \quad (9)$$

$$G_{\text{influence}} : \text{cited evidence participated in the answer path,} \quad (10)$$

$$G_{\text{eval}} : \text{evaluator identity/chronology is intact,} \quad (11)$$

$$G_{\text{access}} : \text{protected-label access telemetry is clear,} \quad (12)$$

$$G_{\text{search}} : \text{search-time contamination is clear.} \quad (13)$$

The claim is promoted only when every required gate passes and the evidence supports it. A resolved negative prerequisite blocks promotion; an unresolved evidential prerequisite leaves the claim undetermined. No failed coordinate can be averaged away by confidence elsewhere. Eight ablations remove one coordinate at a time, including a soft-confidence variant that promotes when at least six of nine gates pass.

5 Methods

5.1 Three-valued outcomes

Every authority decision this paper measures is three-valued rather than binary. A claim may be promoted, blocked, or left *undetermined*. An outcome that could not be determined is recorded and counted separately from an outcome determined to be false, and the two are never merged. The distinction is what makes a missing prerequisite legible: an unresolvable digest, an unreachable checker, or an absent provenance record leaves the claim undetermined, and an undetermined claim is not a refuted one. Throughout the paper an outcome described as *undetermined* is this third value, carried in the evidence record and in every downstream count.

5.2 The implementation under test

The implementation evaluated here is the ORION authority layer; it is the artifact under test, and the rest of the paper calls it the governed pipeline so that each claim reads as a claim about the mechanism rather than about a product. The results tables and figures label its rows and points by that implementation name, since they are projections of the archived campaign record. Source, protocols, and evidence archives are listed in Section 10.

5.3 Prospective freeze and fresh post-repair split

The base protocol was frozen prospectively, and the post-repair campaign reported here is an additive execution binding on top of it, with a fresh hidden split and a repaired panel; Section 10 names the freeze and the exact subject revision. Before any outcome-bearing run, a public-seed preflight exercised 42 hostile authority tests and the complete 420-case pipeline with ten comparator mechanisms, eight ablations, five

Table 1: Protected V2 battery and custody. Hidden labels remained host-only until scored jobs completed.

Slice	Cases	Clean	Hostile	Candidate labels
Post-run public slice	150	30	120	hidden during execution
Protected hostile	120	0	120	hidden
Protected holdout	150	30	120	hidden
Total	420	60	360	hidden

deterministic repeats, protected scoring, typed-panel validation, and an independent reproducer. After the preflight and harness merge were signed and green on the default branch, an independent GitHub-hosted job minted a fresh secret-seeded split. The secret seed was never logged; only a commitment was retained.

The final split contains 420 mechanical-gold cases across 13 families: 60 clean positives and 30 cases in each of 12 hostile/insufficient-evidence families. A 150-case post-run public slice contains 30 clean and 120 hostile cases; distinct protected-hostile and protected-holdout slices remain host-only during scoring. Table 1 summarizes custody.

5.4 Gold-blind execution and custody

The candidate and comparator jobs received exactly two top-level fields per case: an opaque case identifier and the candidate-visible packet. The host mechanically proved that attack family, protected gold, custody class, and expected authority terminal were absent. Candidate and comparator jobs ran in separate GitHub-hosted jobs from the exact subject archive with repository credentials removed. Five deterministic repeats were recorded under file and network syscall tracing. The protected evaluator became eligible to download gold only after both scored lanes completed, and first checked that traces contained neither protected identifiers nor external-IP connections.

The launch workflow also re-queried the signed execution-freeze commit and exact-main CI through the GitHub API, byte-compared the frozen workflow, binding, and harness files to that signed commit, and recomputed host hashes before handing off candidate-visible data. The evaluator and independent reproducer were separate jobs.

5.5 Comparator panel, ablations, and statistics

Seven primary comparator mechanisms mirror the repaired panel: ProvenanceGuard-style source routing, AttributionBench-style multi-source attribution, FIRE-style iterative retrieve/verify, CLAIM-BENCH-style claim evidence, ProvenAI-style citation fidelity/influence, RewardHackingAgents-style contamination/tamper checks, and DeepSciVerify-style evidence escalation. Three auxiliary controls test citation presence, pooled semantic support, and auditability/provenance.

Comparator names therefore use “-style” or describe a mechanism. They are protocol-matched reimplementations under one candidate-visible packet and resource accounting, not executions of the external authors’ software. Every empirical claim below is about the frozen mechanisms in this benchmark, not about published systems in their native environments.

The primary outcome is false authority-promotion rate over gold terminals that do not permit promotion. Clean coverage is promotion rate over the 60 clean positives. H1 requires an absolute false-promotion improvement beyond 0.05 against the strongest frozen comparator; H2 requires clean coverage non-inferior within 0.05; H3 tests superior selection of the undetermined terminal where that is the correct answer. Rate intervals use Wilson 95% intervals; paired effects use a fixed-seed percentile bootstrap. No false promotion, false block, abstention, candidate-caused failure, or clean false negative is removed from its eligible denominator.

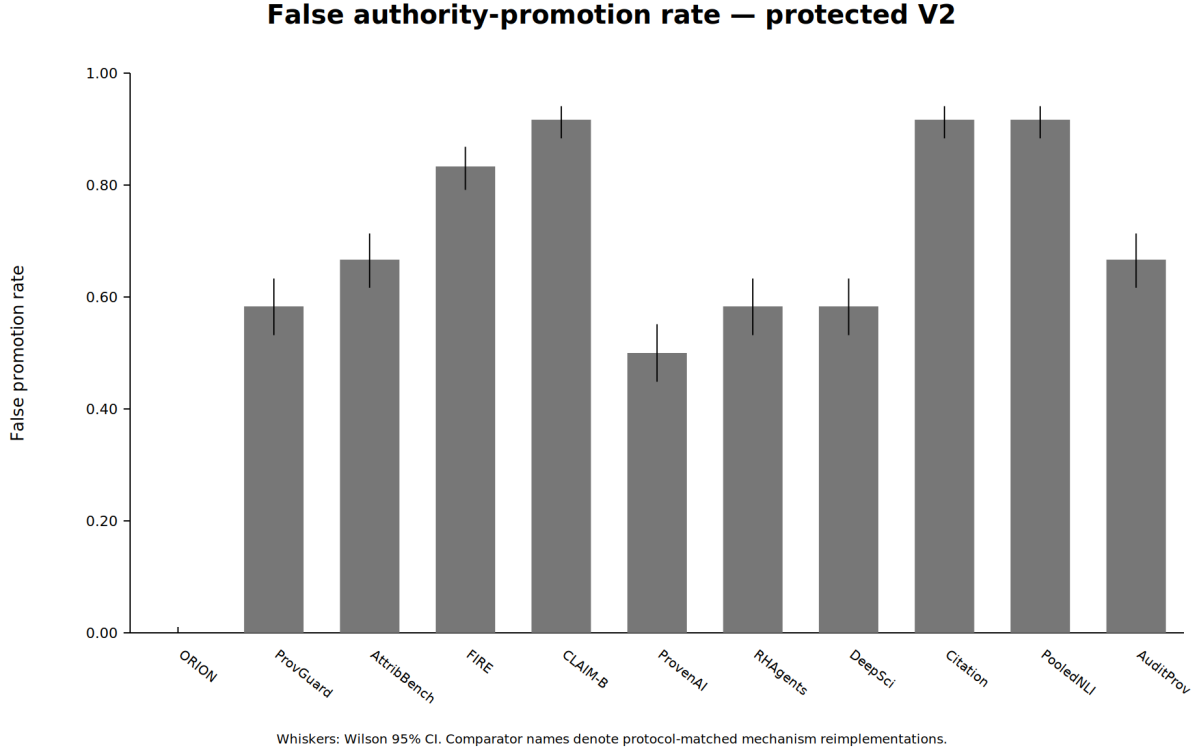


Figure 1: False authority-promotion rate in the protected campaign. Whiskers are Wilson 95% intervals. Comparator labels denote protocol-matched mechanism reimplementations.

6 Results

The typed authority panel passed with no blockers, and a separate implementation independently reproduced the headline counts.

6.1 False promotion and clean coverage

The governed pipeline produced 0/360 false promotions. The strongest frozen comparator mechanism, ProvenAI-style citation-fidelity/influence, produced 180/360 (50.0%). The paired difference was -0.50 with 95% CI $[-0.553, -0.447]$, so H1 passed the predeclared -0.05 practical margin. Figure 1 reports the full mechanism panel.

Both the governed pipeline and the strongest comparator promoted 60/60 clean positives, with zero clean false negatives. The clean-coverage difference and paired 95% interval were both exactly zero, so H2 passed the -0.05 non-inferiority margin. Figure 2 shows that every primary mechanism sits at 1.0 clean coverage; the result therefore does not manufacture a frontier by visually jittering equal values.

On this battery H3 was not supported, and the cause is in the battery rather than in the systems. Every one of the eleven panel members, the governed pipeline included, correctly left all 30 insufficient-evidence cases undetermined, so the difference is zero with 95% CI $[0, 0]$, and the ablations do not instrument the metric at all.

The construction explains why. In this battery the insufficient-evidence family was generated with an *empty* evidence list, while every other family carried exactly one evidence object. The rule “leave the claim undetermined if and only if the evidence list is empty” therefore classifies all 420 cases correctly. Detecting an empty list is not an act of judgement, so the metric was saturated by construction and the comparison

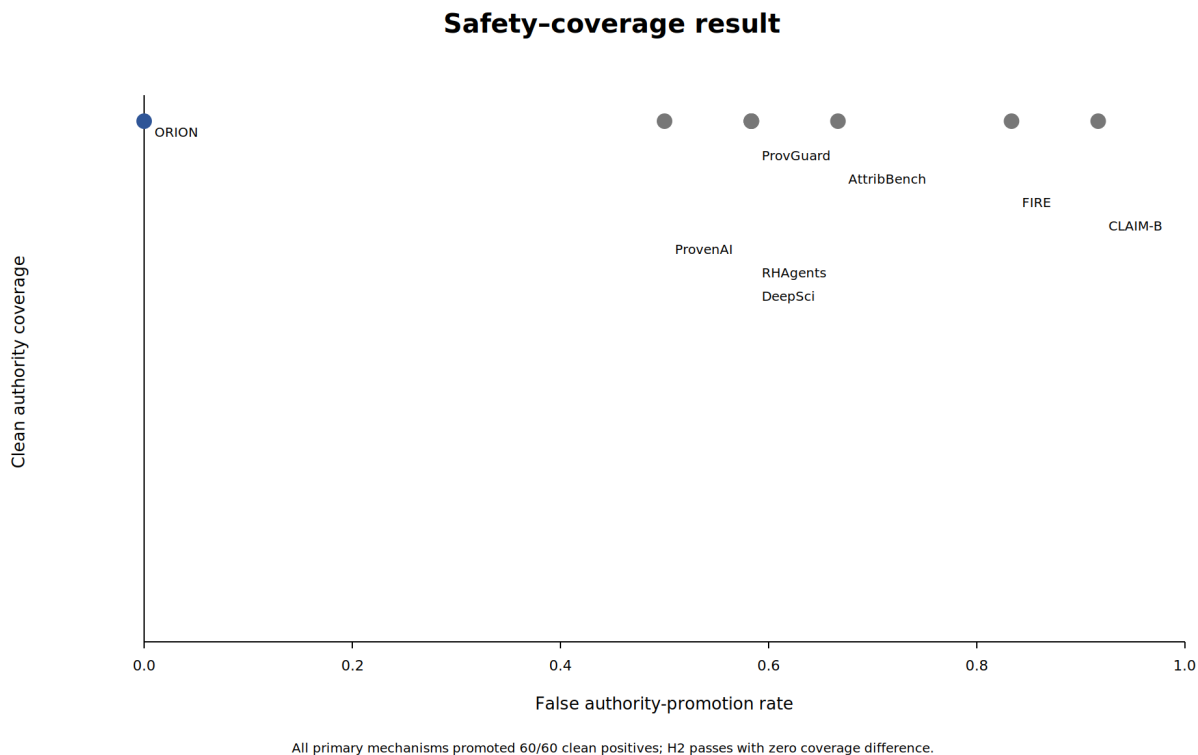


Figure 2: Safety-coverage result. Every primary mechanism promoted 60/60 clean controls, while hostile false-promotion rates differ substantially.

Table 2: Fail-closed outcomes and false-negative cost in protected V2. The only gold family whose correct terminal is ‘undetermined’ contains 30 insufficient-evidence cases.

System	Correctly undetermined	False promotions	Clean FN
ORION	30/30	0/360	0/60
ProvenAI-style	30/30	180/360	0/60

had no resolving power: a difference could not have been observed had one existed. We report this as an inability to discriminate rather than as evidence that no difference exists, and we do not treat the null as a finding about abstention. Table 2 reports abstention and false-negative cost on that basis.

6.2 Repairing the construction, and one run of the frozen panel on it

The first repair changed what must be reasoned about rather than any gold label: cases present evidence that is read but does not establish the claim, in two subtypes, one where the pool contains no supporting record at all and one where a supporting record is present but carries no verifiable provenance. That removed the empty list and did not remove the shortcut. The repaired bodies took five distinct lengths, two of which occur only on undetermined cases, so a rule over evidence-body length recovers the terminal on 20/20 protected cases with no false positives among the other 250 — and the family named itself in English besides. Measured after the fact with the register described below, five of fourteen judgement-free probes recover the undetermined label at informedness 1.00 on that construction, and thirteen of the fourteen do so on the published one. Repairing a named shortcut is not the same as establishing that there is none, so we do not report a panel from it.

The second repair was written against the property rather than against a named cue, and its protocol, seeds and interpretation were frozen in advance of the repair and of any panel outcome. Every case in the battery is shape-identical: one evidence-body template for all 840 records, with the same length, word count, punctuation and character-class profile; fixed container sizes throughout; no absent field anywhere; and support carried by a token that is present on every record, so that only token matching separates support from non-support. The undetermined family is split into two shape-identical subtypes and is a strict subset of a set that also contains 30 blocking cases, so neither “nothing in the pool supports” nor “the assigned record does not support” classifies it. Families differ only in the values of the fields an authority gate is defined over. The identifiability register — fourteen judgement-free probes over counts, container lengths, key shapes, string lengths, character-class and template fingerprints, identifier shapes and scalar values, fitted on the fitting split and scored on the protected one — was run and recorded before the panel was invoked. On the repaired construction it reports informedness 0.0 for every probe on the undetermined axis, on all thirteen registered seeds, against a declared ceiling of 0.0; not one probe ever predicts the undetermined terminal for any protected case.

Run once on that battery, the same frozen panel and statistics report H3 supported. The governed pipeline selects the undetermined terminal on 30/30 eligible cases with 0/360 false promotions; the strongest frozen comparator, selected by H1 exactly as before, selects it on 0/30; the paired difference is 1.0 with 95% CI [1.0, 1.0] against a predeclared threshold of a positive lower bound. The one comparator mechanism with an escalation path, DeepSciVerify-style abstract-to-full escalation, selects it on 15/30, so the margin against that mechanism is 0.5 rather than 1.0; both numbers are needed to read the result. What the difference measures was fixed before the panel ran and is narrower than the headline: terminal expressiveness under a non-compensatory gate lattice — the ability to report an inability — and not a finer-grained scientific judgement. Nine of the ten comparator mechanisms score zero because, once the empty-evidence case is gone, they cannot express an undetermined terminal at all, and a panel of eleven systems in which ten are two-valued cannot separate that reading from a claim about knowing when one cannot check. Clean coverage is 1.0 for all eleven systems on the repaired battery as well, so H2 remains a saturated axis there and is still not a comparative finding. The V2 numbers above are unchanged and are not restated by this run: it is a different battery, and its result does not license reinterpreting the null the V2 construction produced.

6.3 Attack families and ablations

The governed pipeline made no false promotion in any hostile family. The strongest comparator false-promoted all 30 cases in each of six governance/checker families: weak checker, same-lane checker, post-hoc checker/evaluator, search-time contamination, evaluator tamper, and holdout access. It was correct on content substitution, source conflation, wrong-source, pooled-owner, cited-non-influential, and insufficient-evidence families. Figure 3 shows the contrast.

Every registered ablation preserved 60/60 clean coverage while increasing false promotion. Removing exact content binding, checker-lineage independence, hostile-checker discrimination, behavioral influence, or the search-contamination block produced 30/360 false promotions (8.3%) each. Collapsing source/provenance identity produced 60/360 (16.7%); removing evaluator protection/telemetry produced 90/360 (25.0%); and replacing the fail-closed terminal with the predeclared six-of-nine soft-confidence threshold produced 330/360 (91.7%). Table 3 contains the complete panel.

6.4 Attribution, support, cost, and telemetry

Several mechanisms achieve perfect source-attribution accuracy while still false-promoting governance attacks; conversely, pooled semantic-support mechanisms can score well on support while attributing the source incorrectly. Figure 4 separates those coordinates.

Deterministic gate evaluation is slower than the simple mechanism proxies in this synthetic harness: mean latency is about 2.97 ms per case for the governed pipeline versus single-digit microseconds for the primary proxies (Figure 5). This is not an end-to-end serving benchmark; model and network latency are intentionally absent.

False promotion by hostile family

	ORION	ProvenAI
Wrong Source	0%	0%
Content Substitution	0%	0%
Source Conflation	0%	0%
Pooled Support Wrong Owner	0%	0%
Cited Non Influential	0%	0%
Weak Checker	0%	100%
Same Lane Checker	0%	100%
Posthoc Checker Evaluator	0%	100%
Search Time Contamination	0%	100%
Evaluator Tamper	0%	100%
Holdout Access	0%	100%
Insufficient Evidence	0%	0%

ProvenAI-style false-promotes all 30 cases in six governance/checker families; ORION false-promotes none.

Figure 3: False-promotion rate by hostile family for the governed pipeline and the strongest frozen mechanism proxy.

Scored-process telemetry recorded zero protected-identifier hits and zero external-IP connections for both candidate and comparator jobs. Raw traces remain in protected custody. The independent reproducer obtained the same 0/360 versus 180/360 false-promotion counts and 60/60 versus 60/60 clean-promotion counts.

6.5 Exclusion of exploratory evidence

We exclude the repository’s earlier 39-case live-model arm from publication authorization. Post-run audit found internally inconsistent live labels/expected terminals, incorrect opportunity denominators in its summary path, and direct use of a candidate-hidden family label in the live execution path for the governed pipeline and its ablations. Those outputs remain useful diagnostics but are not combined with, substituted for, or used to tune the protected campaign.

Table 3: Protected V2 results. Comparator names denote protocol-matched mechanism reimplementations, not original external software. FP is over 360 hostile opportunities; clean is over 60 clean positives.

System / variant	FP	Clean	Clean FN	Latency (ms)	Resource
ORION	0.000	1.000	0	2.9736	9.0
ProvenanceGuard-style	0.583	1.000	0	0.0037	5.0
AttributionBench-style	0.667	1.000	0	0.0058	3.0
FIRE-style	0.833	1.000	0	0.0069	6.0
CLAIM-BENCH-style	0.917	1.000	0	0.0049	4.0
ProvenAI-style	0.500	1.000	0	0.0034	5.0
RewardHackingAgents-style	0.583	1.000	0	0.0072	5.0
DeepSciVerify-style	0.583	1.000	0	0.0074	6.5
Citation presence	0.917	1.000	0	0.0009	1.0
Pooled support	0.917	1.000	0	0.0049	2.0
Auditability/provenance	0.667	1.000	0	0.0033	5.0
ORION - exact content	0.083	1.000	0	2.7131	9.0
ORION - source/provenance	0.167	1.000	0	2.7077	9.0
ORION - checker lineage	0.083	1.000	0	2.7078	9.0
ORION - hostile checker	0.083	1.000	0	2.7043	9.0
ORION - influence	0.083	1.000	0	2.7045	9.0
ORION - evaluator protection	0.250	1.000	0	2.7116	9.0
ORION soft confidence	0.917	1.000	0	2.4570	9.0
ORION - search block	0.083	1.000	0	2.7087	9.0

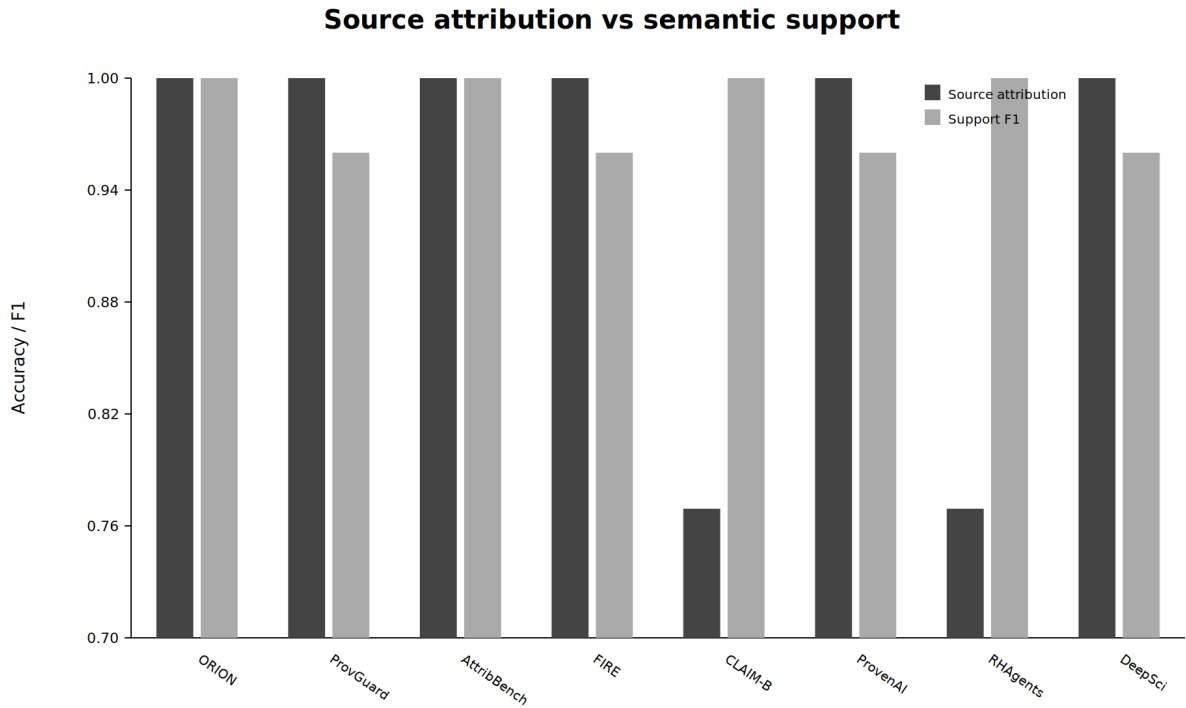


Figure 4: Source-attribution accuracy and support/contradiction F1 for the primary mechanism panel. High values on these coordinates do not guarantee a safe authority terminal.

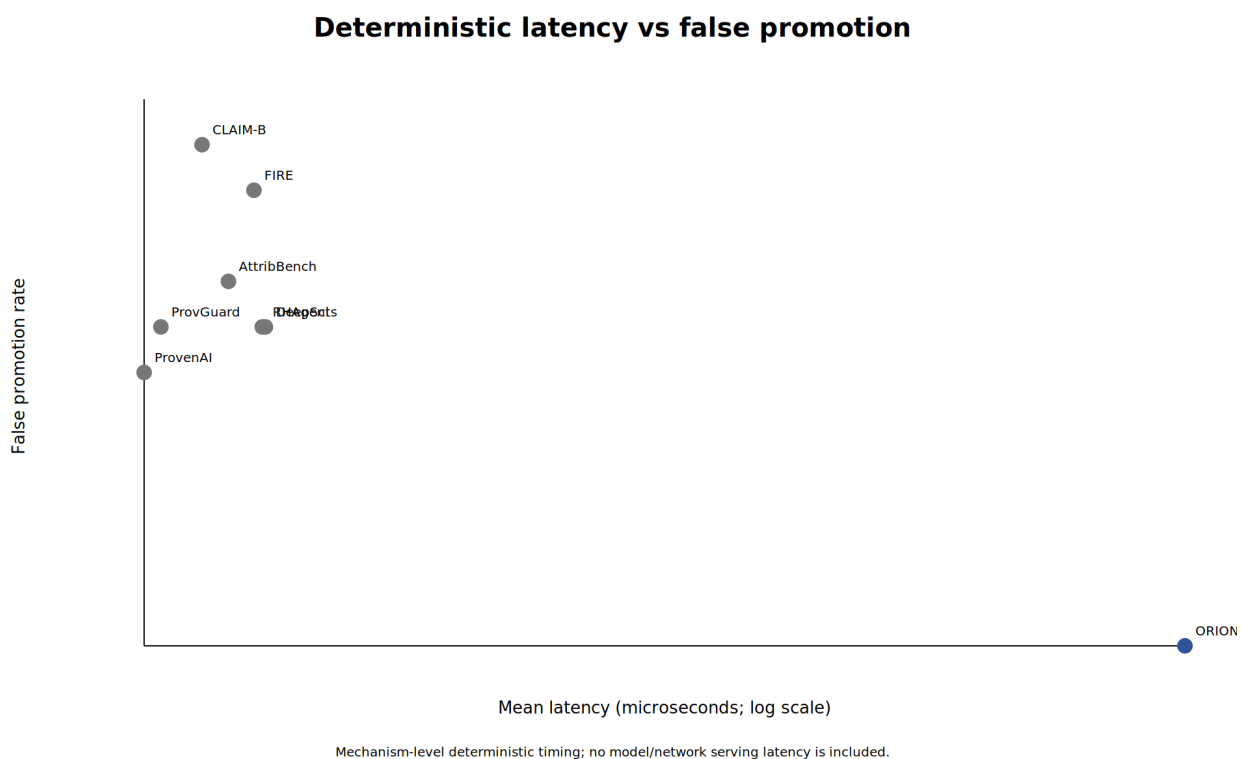


Figure 5: Mean deterministic evaluation latency versus false authority-promotion rate; the latency axis is logarithmic.

7 Post-saturation successor: donor-complete scientific-promotion authority

The protected V2 result asks whether a non-compensatory content/provenance/checker/evaluator pipeline reduces false scientific-authority promotion relative to frozen mechanism proxies. We subsequently ran a separate prospectively frozen successor that raises the comparison standard: ordinary provenance, claim verification, artifact/version binding, evaluator custody, epoch handling and generic authorization are granted to the strong donor product, and the disputed object is only the final *scientific-promotion relation*. The V2 and successor campaigns are separate experiments and are not pooled.

7.1 Donor-engulfment boundary

P4 does not claim source-aware factuality, provenance, claim/evidence verification, citation fidelity, evaluator auditing, contamination defense, abstention or generic authorization as atomic novelties. Those mechanisms are retained as donor-owned local components (Alvarez et al., 2026; Faizan & Alharthi, 2026; Liao, 2026; Tu et al., 2026; Zhou et al., 2026). The successor tests whether successful local verification and generic permission may self-authorize promotion of a target scientific claim.

The P4-X relation is non-compensatory. A target claim may be promoted only when the registered hard scientific-promotion obligations are discharged. In the exact successor battery, the discriminating obligations concern target claim scope, independence of supporting evidence, and unresolved scientific-promotion authority after the ordinary provenance/verification/authorization checks already available to the donor product.

7.2 Exact successor design

The frozen successor contains 400 protected cases spanning five heterogeneous scientific artifact domains, eight hostile/control archetypes and ten protected variants per domain-archetype cell. All systems receive the same candidate-visible information.

The principal arms are:

- **P4-X**: the non-compensatory scientific-promotion relation;
- **B1**: a strong donor-complete generic-authorization product with the ordinary provenance, verification, artifact/version, custody and epoch information, but without the full target scientific-promotion relation;
- **B2**: a compensatory all-signal product that permits successful coordinates to offset failed or unresolved ones;
- **B3**: an ideal typed donor product given every P4-X scientific coordinate and the same promotion predicate; B3 is expected to tie extensionally.

7.3 Protected result

P4-X selected the correct scientific-promotion terminal on 400/400 protected cases. B1 was correct on 250/400, B2 on 50/400, and B3 on 400/400 with zero decision mismatches to P4-X. The paired P4-X minus B1 effect was +0.375 with a domain-stratified bootstrap 95% CI [0.3275, 0.4225]; paired McNemar discordance was 150–0. P4-X made zero false promotions, retained clean promotion at 1.0, and its advantage over B1 had positive direction in every exact domain. An independent implementation that does not import the P4-X execution scorer reproduces the canonical protected-row digest and all arm counts.

The result supports a bounded distinction between local mechanism success and target scientific-promotion authority. In these exact contracts, ordinary provenance, verification, custody and generic permission can all be available while claim scope, evidence dependence or an unresolved scientific obligation still blocks promotion. Conversely, when the donor product receives the same scientific coordinates and the same promotion predicate, it ties exactly; P4 therefore claims no inherent expressivity or centralization advantage.

7.4 Relation to the historical V2 hypotheses

The successor does not retroactively alter the V2 secondary findings. V2 H2 remains a non-inferiority guard in a battery where all primary systems achieved perfect clean coverage. V2 H3 remains not supported/non-discriminating because the historical insufficient-evidence construction was structurally saturated. The P4-X clean controls show that the successor result does not arise from blanket refusal; they do not establish general abstention superiority or difficult-benign recall.

7.5 Claim ceiling

The strongest supported successor statement is therefore architectural but bounded: after strong donor-owned provenance, verification, custody and generic authorization mechanisms are granted, *scientific claim promotion remains a distinct non-compensatory authority relation* in the tested heterogeneous exact contracts. Deployed verifier/provider generality, universal necessity of the exact registered coordinates, general abstention superiority and universal scientific-authority superiority remain outside the evidence.

8 Threat model, limitations, and interpretation

The theory and the evidence have different scopes. Proposition 1 and its factorization corollary hold for any finite exact-terminal decision problem, but they identify what information and interface make a decision attainable; they do not show that a deployed verifier possesses the target-sufficient representation. The empirical layers establish only the representations, relations, and case distributions that were frozen for those layers.

First, the V2/V3 campaigns use 420 synthetic mechanical-gold cases; they do not establish performance on naturally occurring scientific disagreements. The written human-adjudication rubric was never triggered because all final cases were mechanically decidable, so no claim is made about adjudicator agreement. P4-X adds five heterogeneous artifact domains and eight archetypes but remains a 400-case exact-contract study, not a population sample of scientific disputes. Second, scoring is deterministic and uses no LLM provider. It isolates authority-transition semantics but does not measure the error distribution of a full stochastic research agent.

Third, comparator mechanisms are matched reimplementations. They instantiate published ideas under one packet and budget, but the result must not be read as a leaderboard against the original ProvenAI, ProvenanceGuard, DeepSciVerify, FIRE, AttributionBench, CLAIM-BENCH, RewardHackingAgents, or related implementations. Fourth, the clean controls are intentionally unambiguous; 60/60 coverage shows that the governed pipeline did not win by blanket refusal but does not estimate recall on difficult benign cases. Fifth, the hostile taxonomy is finite. Adaptive attacks, compromised host infrastructure, side channels outside retained syscall/network telemetry, and novel evaluator attacks remain outside this campaign.

Sixth, and most consequential for how the three hypotheses should be read, two of them are measured on axes that this battery saturates, and for H3 the mechanism is identified rather than suspected. Clean coverage is 1.0 for all eleven panel systems and for all eight ablations, and the correct-undetermined rate is 1.0 for all eleven systems and is not instrumented in the ablations at all. Only false authority promotion varies: it takes six distinct values across the panel and four across the ablations. The whole empirical separation reported here rests on that one axis. H1 is therefore a measurement; H2’s non-inferiority pass and H3’s null are statements about a battery in which neither quantity was ever observed to move. Testing them would require clean positives that a weaker governance mechanism actually declines, and insufficient-evidence cases that a weaker mechanism actually over-promotes or over-abstains on — cases the current gold construction does not contain. On this battery, therefore, H2 and H3 should be read as design limits rather than as comparative findings.

For H3 such cases now exist, and Section 6 reports them: on the repaired battery the correct-undetermined rate takes three values across the same eleven systems, so the axis moves and the hypothesis is decided by the systems rather than by the construction. Three things that result does not do. It does not retrospectively license the null above — the repaired battery is a different battery, and the numbers reported for the frozen

campaign stand as the record of what that construction produced. It does not widen what H3 claims: nine of the ten comparator mechanisms score zero because they cannot express an undetermined terminal at all, so the quantity measured is terminal expressiveness under a non-compensatory gate lattice, and the margin is 1.0 against the strongest comparator but 0.5 against the one mechanism that can escalate. And it does nothing for H2, whose clean coverage is 1.0 for every system on both batteries; that axis remains a design limit with no case yet built that could move it.

The V3 nuisance audit is also claim-axis specific. Its fourteen frozen probes and thirteen seeds clear the exact CANNOTCHECK axis at informedness 0.0, while the adjudication receipt retains four off-axis residuals and explicitly withholds whole-register clearance. More importantly, nine of ten comparator mechanisms cannot emit CANNOTCHECK. In the terminology of Section 3, V3 has nonzero observed resolution but unmatched terminal attainability. It therefore repairs the V2 construction failure without yet supporting a general scientific-judgement claim.

Seventh, the mechanisms composed here are not claimed as novel, and the boundary is worth stating once and plainly. Provenance tracking, attribution evaluation, citation checking, iterative verification, evidence escalation, contamination detection, evaluator-tamper detection, abstention, benchmark auditing, claim-level auditability, provenance ontologies, and generic behavioral-integrity checking are all prior work, cited in Section 2. The relevance-versus-authorized-evidence distinction is likewise established there. What is measured here is the behavior of the non-compensatory conjunction of those prerequisites under a protected, gold-blind battery, and nothing beyond it.

P4-X does not remove that boundary. It grants the strongest comparison product ordinary provenance, verification, exact artifact/version binding, evaluator custody, epoch handling, and generic authorization before testing the target scientific-promotion relation. Its 400/400 versus 250/400 result shows that the registered generic relation is insufficient on the exact successor contracts; the 400/400 ideal-product tie simultaneously shows that an information-equivalent donor product is sufficient. Neither count licenses a centralization advantage, universal necessity of the exact gate list, or superiority over deployed donor systems.

Recent benchmark-auditing work is an additional caution: expert-built tasks and evaluation harnesses can contain specification, environment, ground-truth, or grading defects that materially alter measured rankings (Tu et al., 2026; Wang et al., 2026a; Kapoor et al., 2025). Deterministic generation, typed-panel validation, exact split/harness binding, protected custody, post-run public slicing, and independent reproduction mitigate this risk, but these controls do not prove that the 13-family taxonomy is complete or that every benchmark-design assumption is correct.

The frozen naturalistic successor, identified as P4.H3.NATURALISTIC.V1, is the upward test rather than a claimed result. It specifies 768 independent source-case-family clusters in 32 strata (four domains by eight unidentifiability mechanisms), pairs every unidentifiable unit with an identifiable same-source control, freezes four CANNOTCHECK-capable primary comparators and a worst-domain gate, and requires independent evaluator/raw-case custody plus a source-disjoint replication panel. Its outcomes have not been accessed and the external panel is missing. No naturalistic, cross-domain, external-comparator, or source-disjoint result is inferred from the protocol or its power calculation.

The appropriate current interpretation is a verification-axis envelope. V2 measures false-promotion separation and records a saturated negative axis; V3 shows that a shortcut-resistant exact axis can still be interface-limited; and P4-X shows both failure of a registered generic-authorization relation and exact equivalence when a donor product receives target-sufficient state and the same predicate. Together they support a fail-closed scientific-promotion interface and the factorization theory that explains its limits. They do not prove universal scientific truth, naturalistic scientific judgement, general abstention ability, or evaluator security.

9 Broader impact and governance

False scientific-authority promotion can amplify incorrect or misattributed claims, particularly when a system presents verification artifacts that appear independent but are not. A fail-closed layer may reduce this risk, but excessive blocking can delay legitimate scientific communication or concentrate power in whoever

controls evidence and evaluator registries. Work on academic integrity and agentic abstention reinforces that refusal can be the correct behavior under genuine insufficiency, while also showing that productive abstention remains difficult and should not be conflated with generic non-response (Yang et al., 2026; Liu et al., 2026). We therefore report abstention cost, clean coverage, custody identity, and artifact transparency rather than optimizing only a security metric.

Protected evaluation creates another trade-off: some labels must remain hidden during scoring, yet publication requires enough evidence for audit. We release safe aggregates and a post-run public slice while retaining protected holdout labels/traces in restricted custody. Exact subject/split/harness hashes and an independent reproduction receipt are published so the result is not represented by an unverifiable scalar score.

10 Data and code availability

Every artifact named anywhere in this paper is listed here with its path, so that no repository location has to be carried in the narrative. Source, protocols, and evidence are version-controlled in the `SzeChunYiu/ORION` repository under `papers/paper-04-verified-scientific-discovery/`; persistent DOI assignment remains pending publication.

10.1 Preregistration

The study was preregistered and frozen before any outcome was accessed, under the identifier `P4.protected-authority.v1`. A reader checking whether a reported analysis was planned or chosen after the fact should compare against that freeze; it is the reason every number reported here can be traced to a decision made in advance of seeing it.

The machine-readable freeze is `protocol/PROTOCOL_V1.json`, the case requirements are `protocol/ATTACK_CASE_SCHEMA_V1.json`, and the statistical rules, metric definitions, and threat model are `protocol/STATISTICAL_ANALYSIS_PLAN_V1.md`, `protocol/METRICS_REGISTRY_V1.json`, and `protocol/THREAT_MODEL_V1.md`. The custody policy and the execution-freeze checklist for the post-repair campaign are `protocol/CUSTODY_POLICY_V1.md` and `protocol/EXECUTION_FREEZE_CHECKLIST_V2.md`. The prospective execution binding, including the exact subject revision and the frozen harness identity, is `protocol/PROTECTED_RUN_BINDINGS_V2.json`. The comparator and ablation disclosure is `host/BASELINE_CONFIGS_V2.json`, and the frozen case generator and verdict-leak check are `protocol/generate_attack_manifest.py` and `protocol/check_verdict_leak_v1.py`.

10.2 Executed campaign and result artifacts

The publication-authorizing campaign is GitHub Actions run 31976589735. The subject under test is commit `f6e51b5c8f905382b8e2f5568d9035fc14241aa1`; the split SHA-256 is `3fe91b669643fa158f2f64c1e6ab70837afbb9b0582e297f1da6e1c3c696fcd9`; the frozen harness SHA-256 is `094f43cb320f8e8e3196049269b20ac22e7e94fa9890b80f27f38ef49f7c82ea`; and the safe publication bundle SHA-256 is `51ac14bc3a6b4b570aaca6d4a41c91f53d9bf2887e66f0620c412f78566a3b44`.

The safe aggregates behind every number in Section 6 are `evidence/protected_v2/PUBLICATION_METRICS_V2.json`; the per-family contrast behind Figure 3 is `evidence/protected_v2/FAMILY_CONTRAST_V2.json`; and the result and custody attestation is `evidence/protected_v2/RESULT_ATTESTATION_V2.md`. Tables 1–3 and Figures 1–5 are regenerated from those two aggregates alone by `figures/generate_figures.py`, which never reads protected per-case labels. The mapping from every result-bearing sentence in this manuscript to the archived artifact that supports it is `evidence/CLAIM_LEDGER_V1.md`. The excluded exploratory arm is documented as `evidence/protected_v2/LIVE_ARM_STATUS.md`; its outputs are retained rather than deleted, and are not pooled with the protected campaign. The nearest-work dispositions that Section 2 discusses in prose are recorded per mechanism in `literature/NEAREST_WORK_AUDIT_2026-08-16.md`.

The safe bundle contains the post-run public attack slice and public per-case verdicts. Protected per-case gold and raw traces are retained separately in restricted custody and are not part of the public release.

10.3 Reproduction

The host-side scoring and independent reproduction paths are `host/evaluate_campaign_v2.py` and `host/independent_reproduce_v2.py`; the latter recomputes the headline counts from the safe bundle without sharing code with the scorer. The binding manifest that ties subject, split, harness, and evaluator identities together is `reproducibility/BINDING_MANIFEST_V2.md`. Reproduction reconstructs the aggregates and the published figures and tables; it does not reconstruct the protected per-case gold, which is what the custody model deliberately withholds.

11 Conclusion

A scientific-verification result requires more than a score. The intended terminal must be identifiable from admissible scientific information, resistant to nuisance decoding, expressible by the compared interfaces, and capable of varying over the panel. The exact-risk and factorization results make these requirements testable: a target relation is implementable precisely when it is constant on the decision representation’s fibres and its terminals are available; mixed fibres impose an irreducible lower bound.

The three P4 evidence layers instantiate different parts of that theory without rewriting one another. V2 eliminated false promotion across 360 hostile opportunities while preserving 60/60 clean promotions, but its H3 family was construction-saturated and remains NOTSUPPORTED. V3 repaired the registered nuisance shortcuts and separated the exact CANNOTCHECK terminal, but its comparison is principally one of interface expressiveness. P4-X then moved the comparator upward: after donor-complete provenance, verification, custody, version, epoch, and generic-authorization mechanisms were granted, the non-compensatory scientific-promotion relation scored 400/400 against 250/400 for the registered generic product and 50/400 for a compensatory product, while an information-equivalent typed product tied at 400/400. The tie is a positive boundary result: scientific promotion need not be centralized; it must be represented with target-sufficient state and the appropriate relation.

The resulting contribution is wider than a gate checklist but narrower than a deployed-system victory. It is a theory of when verification axes can support scientific claims, plus exact protected evidence that local verification and generic permission need not suffice for scientific promotion. The frozen 768-cluster naturalistic successor is the next discriminator and currently has no external-panel outcome. Naturalistic, verifier-provider, source-disjoint, and universal scientific-authorization superiority therefore remain open research targets, not promoted claims.

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A Freeze and custody identifiers

The final hidden set was created only after the public preflight and harness merge were signed and green. The independent host run is 31975937488; the protected attack-set SHA-256 is 6b8b6243040163dfa582dbeb8896e2c68d6b263e3227b64e7bab199d8599897c; and the candidate-manifest SHA-256 is c5df341a6bf63f83d2af17492c498d4d36a944ad6616097a630d8390b7bda331. Execution-freeze merge 99bcacc82224089c34019ad82287754388dadbc5 was GitHub-verified before launch. The launch workflow independently re-queried that signature and exact-main CI and byte-compared frozen workflow, binding, and harness files against the signed freeze commit before handoff.

B Ablation interpretation

Five 8.3% ablations each expose a hostile family governed by the removed coordinate. Source/provenance collapse exposes two families (16.7%), evaluator-protection removal exposes three (25%), and soft confidence permits promotion in eleven of twelve hostile families (91.7%). This monotone pattern is descriptive evidence that the final zero-false-promotion result is not generated by one redundant gate alone.